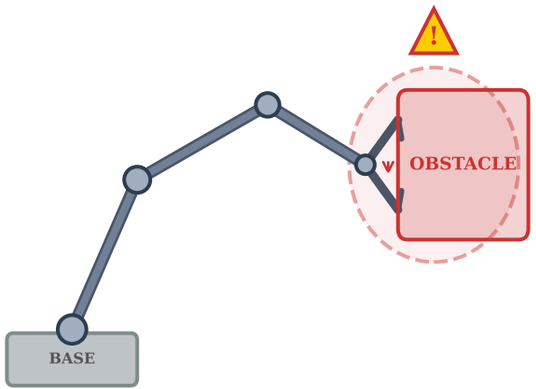
(a) The Problem

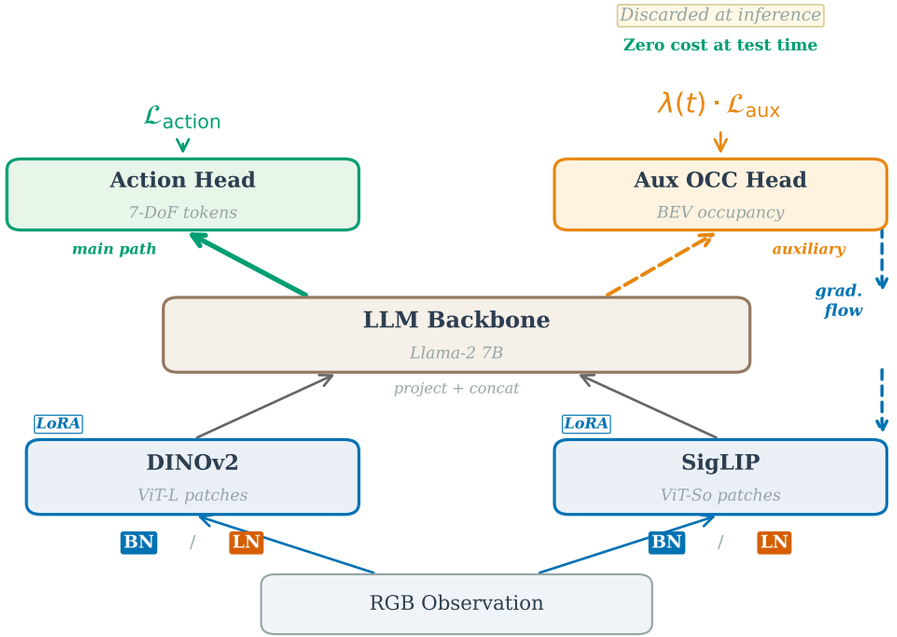
VLA policies lack collision awareness

Deployed in safety-critical environments where collisions cause irreversible damage

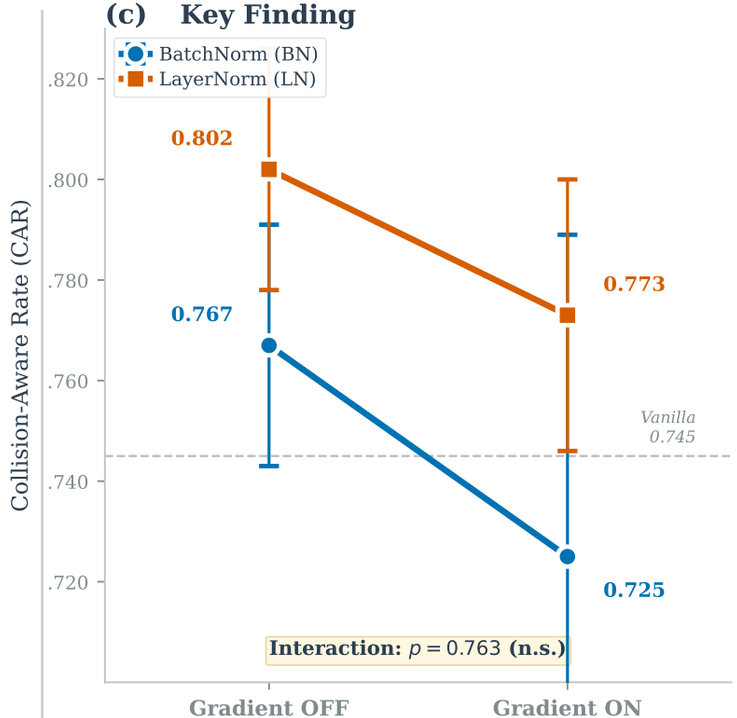


SafeLIBERO benchmark: 10 tasks, collision-aware evaluation

(b) Our Approach



(c) Key Finding



Gradient flow and norm type act additively — LN consistently safer, gradient OFF consistently safer